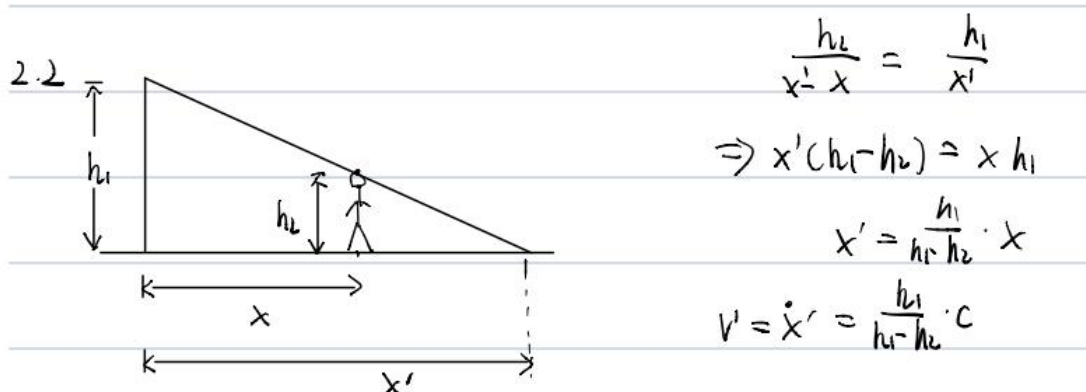


第二章作业



2-6 题解:

$$\rho = l e^{\varphi k}$$

$$(1) \quad v_\rho = l e^t, \quad v_\varphi = \frac{l}{k} e^t$$

$$a_\rho = l e^t \left(1 - \frac{1}{k^2}\right), \quad a_\varphi = 2 \frac{l}{k} e^t$$

$$\rho = l(1 + \cos \varphi)$$

$$v_\rho = -l\omega \sin \omega t, \quad v_\varphi = l\omega(1 + \cos \omega t)$$

$$(2) \quad a_\rho = -l\omega^2(1 + 2\cos \omega t)$$

$$a_\varphi = -2l\omega^2 \sin \omega t$$

$$\rho = l\sqrt{2\cos 2\varphi}$$

$$v_\rho = \frac{-l \sin t}{\sqrt{2\cos t}}, \quad v_\varphi = \frac{l}{2} \sqrt{2\cos t}$$

$$(3) \quad a_\rho = -l(1 + 2\cos^2 t) (2\cos t)^{-3/2},$$

$$a_\varphi = -l \sin t / \sqrt{2\cos t}$$

$$\rho = l(\sin \varphi \cdot \operatorname{tg} \varphi - \cos \varphi)$$

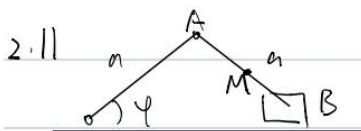
$$(4) \quad v_\rho = \frac{lt(t^2 + 3)}{(t^2 + 1)^{3/2}}, \quad v_\varphi = \frac{t^2 - 1}{(1 + t^2)\sqrt{t^2 + 1}}$$

$$a_\rho = \frac{4l(1 - t^2)}{(t^2 + 1)^{5/2}}, \quad a_\varphi = \frac{8lt}{(1 + t^2)^{5/2}}$$

T2.9

由于 $\vec{a} \perp \vec{v}$, 故 $\vec{a}_v = 0$, $\vec{a} = \frac{v^2}{\rho} \vec{n} = c(\vec{v} \times \vec{r})$

取绝对值: $\rho = \frac{v^2}{c |\vec{v} \times \vec{r}|}$



$$x_m = \frac{3}{2} a \cos \theta \quad y_m = \frac{a}{2} \sin \theta$$

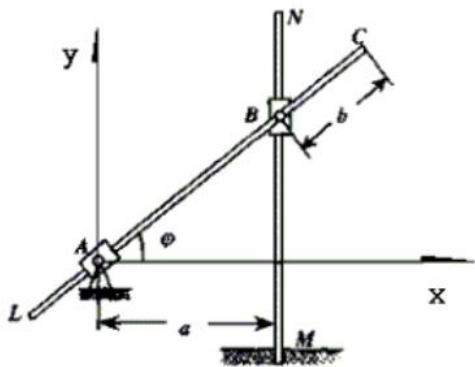
$$\Rightarrow V_{x,m} = -\frac{g}{2} \alpha \sin(\omega t) \cdot \omega \quad V_{y,m} = \frac{g}{2} \alpha \cos(\omega t) \cdot \omega$$

$$X_B = 2a \cos \varphi$$

$$\Rightarrow V_B = -2a \sin \omega t \cdot \omega.$$

2-15 题解:

建立图示参考坐标系 Axy 。写出 C 点的运动规律为:



$$x_c = b \cos \varphi + a = b \cos \omega t + a$$

$$y_c = b \sin \varphi + atg \varphi = b \sin \omega t + atg \omega t$$

C 点的速度 v_x , v_y 分别为:

$$v_x = \dot{x}_c = -b\omega \sin \omega t$$

$$v_y = \dot{y}_c = b\omega \cos \omega t + a\omega / \cos^2 \omega t$$

C 点的加速度 a_x , a_y 分别为:

$$a_x = \ddot{x}_c = -b\omega^2 \cos \omega t$$

$$a_y = \ddot{y}_c = (-b\omega^2 + 2a\omega^2/\cos^3 \omega t)\sin \omega t$$

2-20 题解:

$\therefore A, B$ 两点在垂直于 AB 连线的速度分量是相等的。

$\therefore AB$ 之间的连线始终平行于原来的连线。

A 沿直线运动. 即 φ_1 恒定 (V_1, V_2 恒定)

又 $\therefore V_1 \sin \varphi_1 = V_2 \sin \varphi_2 \quad \therefore \varphi_2$ 恒定

即 B 也一定沿直线运动.

$$B \quad \begin{cases} x_B = V_2 \sin \varphi_2 \cdot t \\ y_B = V_2 \cos \varphi_2 \cdot t \end{cases} \quad t=0 \text{ 时 } s_B = 0$$

$$\therefore \frac{ds_B}{dt} = (\dot{x}_B^2 + \dot{y}_B^2)^{1/2} = V_2 \quad \therefore s = \int_0^t V_2 dt = V_2 t$$

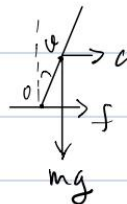
对于 $A \quad \begin{cases} x_A = V_1 \sin \varphi_1 t \\ y_A = V_1 \cos \varphi_1 t \end{cases}$

$$\therefore AB = r_0 + y_A - y_B = r_0 + V_1 t \cos \varphi_1 - V_2 t \cos \varphi_2$$

$$AB = 0 \text{ 时相遇 此时 } y_B = r_0 + y_A$$

$$\text{即 } V_2 t \cos \varphi_2 = V_1 t \cos \varphi_1 + r_0$$

$$\therefore t = \frac{r_0}{V_2 \cos \varphi_2 - V_1 \cos \varphi_1}$$

2.24  $a = \frac{v^2}{r} \quad f = m \frac{v^2}{r} \in \mu mg$

$$\Rightarrow v \leq \sqrt{\mu g r} = \sqrt{0.4 \times 9.8 \times 10} = 14 \text{ m} \cdot \text{s}^{-1}$$

相对 O 点的力矩平衡: $mg \cdot L \sin \theta = \mu mg \cdot L \cos \theta$

$$\Rightarrow \tan \theta = \mu = 0.4 \Rightarrow \theta = \tan^{-1}(0.4) \approx 21.8^\circ$$

2-35 题解

解。当小球以高为 H 处滑下时, 小球在 D 点的速度为 v

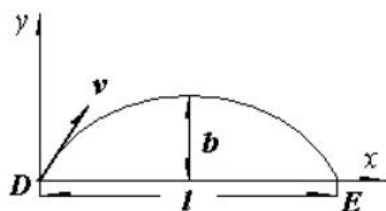
则由动能定理 $mgH = \frac{1}{2}mv^2$ (1)

$\therefore$ 要求最小高度 H , 应使 v 最小而使小球正好落在 E 点,

以 D 为原点, $\overline{DE}$ 为 x 轴建立坐标系如图示。

由几何关系, $\vec{v}$ 与 x 轴的夹角为 45°

$$\therefore \vec{v} = \frac{\sqrt{2}}{2}v(\vec{i} + \vec{j})$$



设经 t 秒由 D 到 E , 则 $\frac{t}{2}$ 时 $v_y = 0$, $\Rightarrow \frac{\sqrt{2}}{2}v = g \frac{t}{2}$ (2)

$$l = \frac{\sqrt{2}}{2}vt \quad (3)$$

$$b = \frac{1}{2}g\left(\frac{t}{2}\right)^2 \quad (4)$$

连立求解 (1), (2), (3), (4) 可得

$$H_{\min} = \frac{1}{2}l = 20 \text{ cm}$$

$$h = b = \frac{1}{4}l = 10 \text{ cm}$$